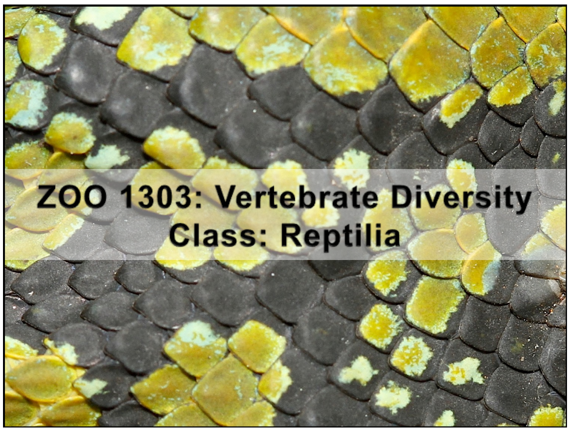
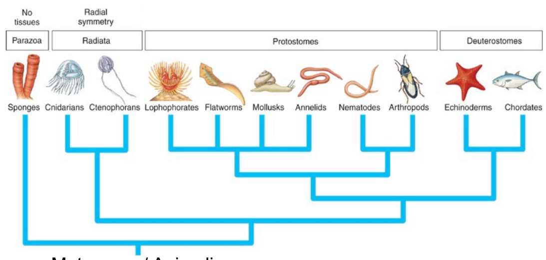


ZOO 1303: Vertebrate Diversity
Class: Reptilia

Class: Reptilia

Domain: Eukarya/ Eukaryota
Kingdom: Animalia
Phylum: Chordata
Subphylum: Vertebrata
Class: Reptilia

Phylogeny of Metazoa/Kingdom Animalia



Phylum: Chordata

Ancestral (Autapomorphic) characteristics of the members of the phylum chordata

- Bilateral symmetry
- Segmented body
- Three germ layers
- Well developed coelom
- Segmented muscles in an unsegmented trunk
- Ventral heart, with dorsal and ventral blood vessels; closed blood system
- Complete digestive system

Phylum: Chordata

Derived (Apomorphic) characteristics of the members of the phylum chordata

- Dorsal, tubular nerve cord
- Notochord present at some stage in the life cycle
- Pharyngeal slits present at some stage in the life
- Postanal tail, projecting beyond the anus at some stage
- Endostyle/ Thyroid

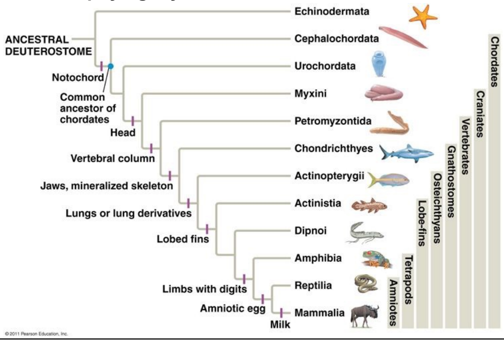
Phylum: Chordata

Derived (Apomorphic) characteristics of the members of the subphylum vertebrata

- Integument- two layers: epidermis (stratified epithelium) and dermis (connective tissue)
- Cartilaginous or bony endoskeleton
- Many muscles attached to endoskeleton
- Excretory system – paired kidneys
- Highly differentiated brain; 10 or 12 pairs of cranial nerves
- Endocrine system
- Body plan consisting typically of head, trunk, and postanal tail

Phylum: Chordata

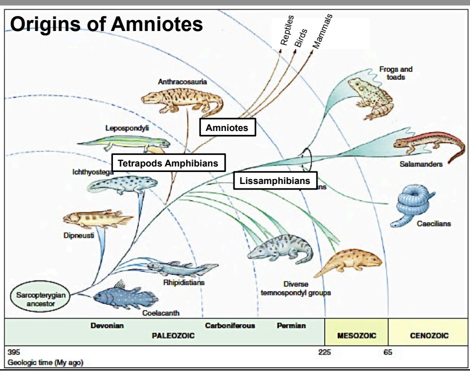
Chordate phylogeny



Class: Reptilia

- Commonly called reptiles
- Reptiles – amniotic tetrapod vertebrates
- Earliest amniotes evolved in the Carboniferous period (360 - 300 Mya) from amphibians

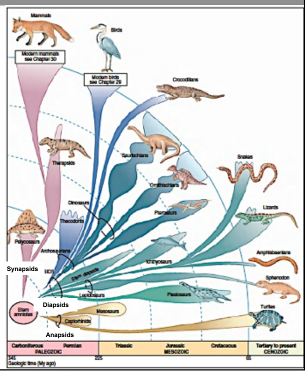
Class: Reptilia



Class: Reptilia

Origins of Amniotes

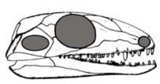
- By the late Permian period (~300 MYA), amniotes had separated into two lineages
  - Synapsids
  - Diapsids



Class: Reptilia

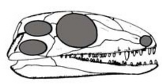
Origins of Amniotes

Synapsids - single pair of temporal openings behind the orbits in skull



Found in the present day mammals

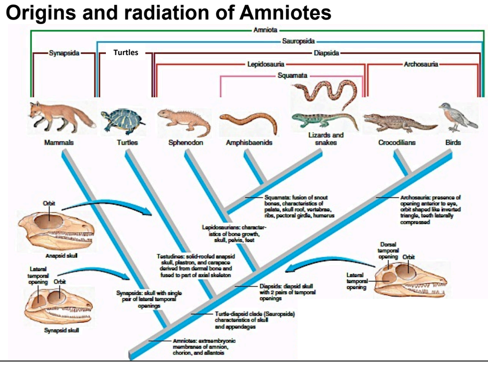
Diapsids - two pairs of temporal openings behind the orbits in skull



Found in the present day, turtles crocodiles, birds & all other reptiles

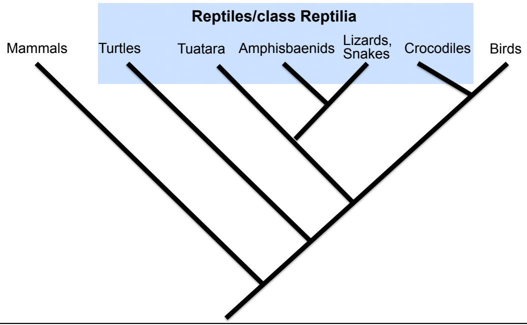
Class: Reptilia

Origins and radiation of Amniotes



Class: Reptilia

Origins and radiation of Amniotes



Class: Reptilia

Biological Innovations

1) Shelled, **amniotic** egg
   - Supplied with extra-embryonic membranes
   - provide a complete life-support system for the enclosed embryo
   - allowed amniotes to lay eggs on land

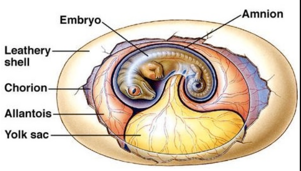
- Shelled (**amniotic**) egg - contains food for embryonic development

Class: Reptilia

Biological Innovations

1) Shelled, **amniotic** egg

- **Amnion**- encloses the embryo in a fluid-filled cavity
- **Allantois** - chamber for the storage of nitrogenous wastes, respiratory surface
- **Chorion** – O_2 and CO_2 freely pass through
- Outermost membrane – porous or leathery shell
- Yolk sac - provides nutrients for the developing embryo

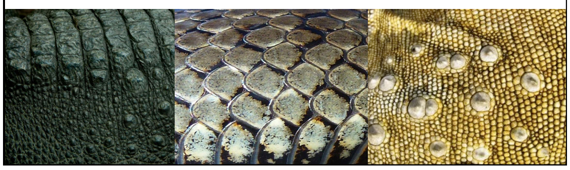


Class: Reptilia

Biological Innovations

2) A **tough, dry, heavily keratinized skin -** provides protection against desiccation and injury

   Scales in reptiles and feathers in birds arise as epidermal elevations overlying a nourishing dermal layer



Class: Reptilia

Biological Innovations

3) Larger and stronger jaw muscles - permit **powerful jaw closure**

   Temporal openings in the diapsid skull provide space for bulging temporal muscles

Class: Reptilia

Biological Innovations

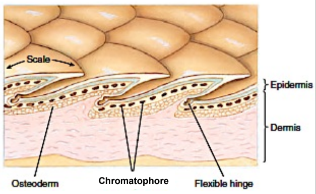
4) Internal fertilization
   Sperm introduced directly into the female reproductive tract with a copulatory organ

5) Effective **adaptations for water conservation -**
   Kidney that excretes nitrogenous wastes as uric acid
   - allowed reptiles (and birds) to reduce water loss and occupy many terrestrial habitats

Class: Reptilia

Characteristics of Reptiles
Skin

- Two layers in skin
  - Epidermis
  - Dermis
- Epidermis, shed periodically
- **Scales** are formed of keratin



Class: Reptilia

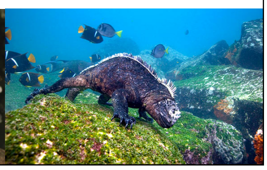
Characteristics of Reptiles
Reproduction

- Sexes separate, sexual reproduction (a few exceptions)
- Copulatory organ for internal fertilization
- Sperm must reach the egg before the egg is enclosed in membranes
- Sperm from the paired testes are carried by the vasa deferentia to the copulatory organ
- Female system consists of paired ovaries and oviducts


Class: Reptilia

Characteristics of Reptiles
Feeding

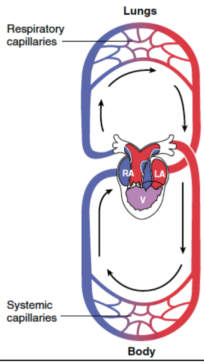
- Jaws are efficiently designed for applying crushing or gripping force to prey
- Larger and longer jaw muscles
- Mainly carnivores
- Some lizards are omnivores or herbivores

Class: Reptilia

Characteristics of Reptiles
Circulatory system

- Comparatively efficient circulatory system and higher blood pressure
- Three chambers in heart – Ventricle is incompletely separated
- Little mixing of oxygenated and deoxygenated blood
  Two almost functionally separate circulations
- Crocodilians have two completely separated ventricles

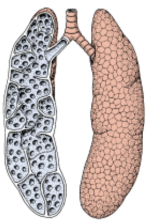


Class: Reptilia

Characteristics of Reptiles

Respiration

- Comparatively well developed lungs
- Depend exclusively on lungs for gas exchange
- Lungs are subdivided into numerous interconnecting air sacs that are highly vascularized
- Air is taken into the lungs by enlarging the thoracic cavity (Negative pressure mechanism)



Class: Reptilia

Characteristics of Reptiles
Excretion

- Evolved efficient strategies for water conservation
- Nephrons lack the loop of Henle - unable to produce liquid urine more concentrated than their body fluid
- Nitrogenous waste - uric acid, water insoluble
- Salt glands (located near the nose or eyes) – secretes excess salts

Class: Reptilia

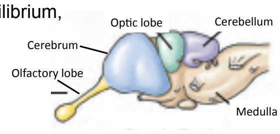
Characteristics of Reptiles
Movement on Land

- Have good body support and more efficiently designed limbs to travel on land
- Some have secondarily lost limbs

Class: Reptilia

Characteristics of Reptiles
Nervous & Sensory system

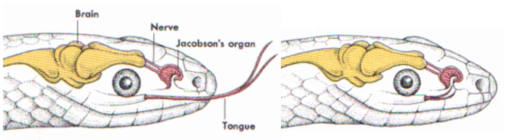
- Considerably complex
- Cerebrum larger than in amphibians
- Crocodilians have the first true cerebral cortex – allows complex behaviour
- Medulla - controls heartbeat, respiration, vascular tone, gastric secretions, and swallowing
- Cerebellum- controls equilibrium, posture and movement



Class: Reptilia

Characteristics of Reptiles
Nervous & Sensory system

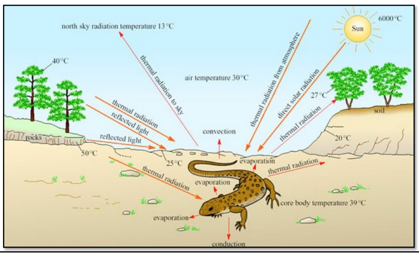
- Eyes - retinas have cone and rod cells
- Lizards – hearing not significantly developed (exception geckoes)
- Jacobson's organ - specialized olfactory chamber in the buccal cavity



Class: Reptilia

Characteristics of Reptiles
Thermoregulation

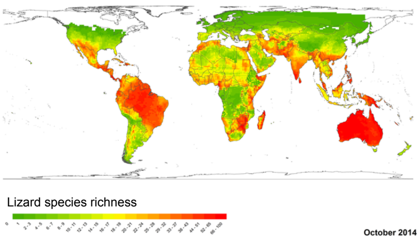
- Ectothermic
- Behavioural thermoregulation



Class: Reptilia

Characteristics of Reptiles
Distribution

Distribution of global reptilian diversity: ~ 11,570 species (Reptile database: http://reptile-database.reptarium.cz/ 2021)



Class: Reptilia

Characteristics of Reptiles
Distribution

Distribution of global reptilian diversity: ~ 11,130 species (Reptile database: http://reptile-database.reptarium.cz/ 2020)